

Manav Bhullar

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EDUCATION

Thapar Institute of Engineering and Technology

2023 – 2027

B.E. in Computer Engineering

- **Relevant Coursework:** Operating Systems, Object-Oriented Programming, Database Management Systems, Computer Networks, Software Engineering

ACHIEVEMENTS & CERTIFICATIONS

Solved **400+** **Data Structures & Algorithms** problems across LeetCode and GeeksforGeeks

Coursera: Advanced Relational Database and SQL, APIs in Node.js: RESTful Backend Development, CI/CD Pipeline with Docker

EXPERIENCE

Virsa Society, Thapar Institute

2023 – 2025

Technical Head (Core Member, 2023 – 2024)

- Approached 30+ local vendors in Patiala for stall partnerships at Virsa Mela, converting 20 into confirmed sponsors for an event drawing 4,000+ student attendees
- Rebuilt the society website (ssavirsa.in) frontend from scratch in **React**, migrating off a static HTML/CSS setup, and independently built the events and gallery pages

PROJECTS

Real-Time Ride-Matching Engine (Floq) | *Node.js, PostgreSQL, Redis, Socket.io* | Live

- Managed ride lifecycles utilizing a **Finite State Machine (FSM)**; prevented double-matching race conditions using Redis SET NX PX distributed locks (with Lua atomic release) and PostgreSQL FOR UPDATE SKIP LOCKED row-level locks
- Implemented a **read-through cache** pattern using Redis to optimize PostgreSQL read latency, and leveraged a **Pub/Sub** architecture via Socket.io for real-time event-driven dispatching and location updates
- Built a ride-matching engine utilizing a **constrained backtracking algorithm** to generate pickup/drop-off permutations, pruning the search space from 40,320 to 2,520 for 4 riders with a 30% max detour constraint
- Optimized multi-rider connectivity validation by replacing higher-order array allocations with single-pass manual loops and Int32Array; benchmarked a **4.8x speedup** (34.3ms to 7.1ms for 10k iterations) using process.hrtime
- Wrote **108 integration tests** covering FSM state transitions, Pub/Sub cascade cancellations, read-through cache invalidation timing, and identical-coordinate reuse attacks with a 100% pass rate

Campus Marketplace (Second-Hand Platform)

- Built a full-stack marketplace enabling students to buy and sell second-hand products, with **Google OAuth + JWT authentication** and protected routes
- Designed **RESTful APIs** for product and user management; deployed frontend and backend independently for scalability

Housekeeping Management System

- Built a **MERN**-based system for managing service requests and workflows, with structured database schema and **CRUD APIs**

Medicine Price Comparator

- Built a medicine comparison platform with **JWT authentication**, **RESTful APIs**, and a favorites feature

TECHNOLOGIES

Languages: JavaScript, C++

Web: React.js, Node.js, Express.js, Socket.io, HTML, CSS

Database: PostgreSQL, MongoDB, Redis, Prisma ORM

Tools & Testing: Git, GitHub, Postman, Docker, Jest, Vercel, Render